

RESEARCH INTERESTS

Computer Vision, Adversarial Attacks, Continual Learning, Natural Language Processing, Model Steering and Deep Learning. I have also started exploring Complexity Theory.

EDUCATION

Islamic University of Technology (IUT)

BSc in Computer Science and Engineering

 Gazipur, Bangladesh

 Jan 2020 – Jun 2024

- ▶ CGPA: **3.88/4.00**, ranked **9th** in the class. Studied on the OIC scholarship.

Rajuk Uttara Model College

Higher Secondary Certificate (HSC), Dhaka Board. GPA: 5.00/5.00

 Dhaka, Bangladesh

 2019

Shahajuddin Sarker Model School and College

Secondary School Certificate (SSC), Dhaka Board. GPA: 5.00/5.00

 Tongi, Gazipur

 2017

RESEARCH EXPERIENCE

A Handful of Patches: The Low-Rank, Decoupled Vulnerability of NR-IQA

 Jul 2025 – Present

Reaz Hassan Joarder, Al-Mubin Khan Nabil, Irfan Nafiz Shahan, **Md. Irfanur Rahman Rafio**, Md Hasanul Kabir

Work in progress

 [OpenReview](#)

- ▶ Studying whether the adversarial vulnerability of no-reference image quality assessment (NR-IQA) models, now used as losses and quality filters, is shared across images and models.
- ▶ Constrained attack: per-image coefficients over a small global dictionary of learned 16×16 patch atoms, shared across all images and models, under an $L_\infty \leq 8/255$ budget.
- ▶ Across HyperIQA (CNN), MUSIQ (transformer) and CLIP-IQA+ (vision-language), **four learned atoms recover 63–98%** of the PGD-achievable score change in both directions, well beyond a random dictionary of the same size.
- ▶ Within the same basis, one perturbation realizes **all 27** raise, lower and hold targets across the three models, while held models drift by less than 0.01. The vulnerability is low-rank and shared, yet decoupled across models.

TEACHING EXPERIENCE

Islamic University of Technology (IUT)

Part-time Lecturer, Department of CSE

 Gazipur, Bangladesh

 Sep 2024 – Present

- ▶ Teach Structured Programming II, Pattern Recognition, Algorithms and Artificial Intelligence labs.
- ▶ Redesigned **Pattern Recognition Lab** to turn the black boxes of deep learning into white boxes. Students complete a small PyTorch-like library with a hand-written autograd Tensor before moving to PyTorch, and study a CNN with hand-assigned weights.
- ▶ Restructured Structured Programming II Lab around recursion, complexity, Git and memory, with lectures animated in Manim.
- ▶ Created a modular $\text{\LaTeX}$ template for lab assignments, now used by every programming lab in the department, and an offline **local judge** with test generation and stress testing, adopted by other labs.
- ▶ Planned and initiated a uniform OS image for the labs, which every lab at IUT now runs on.

Competitive Programming Trainer

 Jan 2023 – Present

- ▶ Started teaching weekly classes for juniors in January 2023 and was officially appointed as the trainer in September 2024.
- ▶ Teach data structures, algorithms, problem solving and contest strategy, and lead post-contest analyses.

United International University (UIU)

Lecturer, Department of CSE

📍 Dhaka, Bangladesh

📅 Nov 2024 – Feb 2026

- ▶ Taught Structured Programming Language, Object Oriented Programming, Data Structures and Algorithms I, and Data Visualization, along with their labs.
- ▶ Designed the **first offering of Data Visualization** and its lab, including the syllabus, lectures, lab tasks and exams.
- ▶ Added a lecture on binary search to Data Structures and Algorithms I, with a practice problem bank.
- ▶ Prepared solutions and marking rubrics after every exam for students and fellow teachers.
- ▶ Moderated and trained the UIU Competitive Programming Community.

Shahajuddin Sarker Model School

Voluntary Math Instructor

📍 Tongi, Gazipur

📅 May 2022 – Jul 2022

- ▶ Taught free math classes during a semester break, encouraging reasoning over memorization.

HONORS AND AWARDS

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|--|------------|
| ▶ Ranked 9th in the IUT CSE class of 2024 (CGPA 3.88/4.00) | 2024 |
| ▶ 5th of 309 teams, ICPC Asia Dhaka Regional Contest 2024 ; first to solve Problem C | 2024 |
| ▶ Qualified for the ICPC Asia West Continent Final Contest | 2023, 2024 |
| ▶ Qualified for the Rayan 2025 World Finals , Sharif University of Technology | 2024 |
| ▶ Codeforces Expert, maximum rating 1892 ; 1200+ problems solved across online judges | |
| ▶ Highest individual points, IUT CodeRush 1.0 Capture The Flag (team placed 2nd) | 2023 |
| ▶ OIC Scholarship, IUT admission test | 2019 |

INDUSTRY EXPERIENCE

Therap (BD) Ltd.

Associate Software Engineer (Development)

📍 Dhaka, Bangladesh

📅 Jul 2024 – Oct 2024

- ▶ Joined through campus recruitment; trained in object oriented design, Java, JDBC, Hibernate and JSP.

BinduLogic Limited

Software Engineer Intern

📍 Dhaka, Bangladesh

📅 Jun 2023 – Sep 2023

- ▶ Built the backend of BinduHealth OPD in TypeScript with Prisma, including the database schema and request validation.
- ▶ Led an improvement of the script checker in BinduExam, using graph algorithms and geometry to make it faster and more accurate.

PROBLEM SETTING AND JUDGING

Set, tested or coordinated **16 programming contests** and wrote **28 original problems**, including testing [Codeforces Round 1067 \(Div. 2\)](#) and coordinating the Intra-IUT Junior Programming Contests and the IUT Intra University Programming Contest 2025. My favorite problems are **pure ad-hoc problems** that reward out-of-the-box thinking, followed by math problems.

Bangladesh Association of Problem Setters (BAPS)

Problem Setter and Judge

📍 Bangladesh

📅 Dec 2024 – Present

- ▶ Wrote 4 original problems and tested 14 for the **ICPC Asia Dhaka Regional 2025** and its preliminary.
- ▶ Built the plagiarism checker for the ICPC Dhaka Regional 2025 Preliminary, which flagged 251 submissions from 124 teams across 51 institutions.
- ▶ Set and tested problems for the KUET, UIU and NSU inter-university programming contests.

Bangladesh Competitive Programming Society (BCS)

Problem Setter and Judge

📍 Bangladesh

📅 Apr 2025 – Present

- ▶ Set and tested problems for the DUET, CUET and SUST inter-university programming contests.
- ▶ Co-coordinated IUT IUPC 2026, finalized its problemset with Shahjalal Shohag and wrote 4 of its problems.

SELECTED PROJECTS

IUPC Rating and Slot Allocation System [🔗 Code](#)

📅 Jan 2025 – Present

🔗 *Python, pandas, NumPy*

Open-source system that rates Bangladeshi universities by their teams' results and allocates slots for inter-university contests. Almost every national IUPC has used it since 2025, including hosts outside BCS.

Mini Deep Learning Library

📅 Jun 2026

🔗 *Python, NumPy, Matplotlib, Jupyter*

A small PyTorch-like library for teaching, with a Tensor class and automatic differentiation, models, losses, metrics and decision-region plots. Students use it to build k-NN, linear and neural network classifiers.

Local Judge for Algorithms Lab [🔗 Code](#)

📅 May 2025

🔗 *Bash, Python, Linux*

Offline judge that compiles and runs C, C++, Python and Java submissions against test data with time and memory limits. The full version also generates test cases and stress-tests solutions.

GPT (Goru Party T-shirt) [🔗 Code](#)

📅 Jan 2024 – May 2024

🔗 *TypeScript, Bun, Express, Prisma, PostgreSQL, React*

A platform where IUT CSE-19 students proposed and voted on each other's nicknames for their class T-shirts. Set up the project and built the database schema and several backend modules.

RapidRoll Advanced [🔗 Code](#)

📅 Oct 2021

🔗 *C++, SFML*

A tribute to Rapid Roll, the classic phone game from our childhood, rebuilt with more platforms and bonuses.

Manim Slides [🔗 Code](#)

📅 Apr 2026 – Jun 2026

🔗 *Python, Manim, Bash*

Animated lecture slides for the Structured Programming II Lab at IUT, built on a small reusable presentation library I wrote on top of Manim.

TECHNICAL SKILLS

- ▶ **Languages:** C/C++, Python, TypeScript, JavaScript, Java, Bash, MATLAB
- ▶ **Data Science:** PyTorch, scikit-learn, NumPy, pandas, OpenCV, Matplotlib, Jupyter, web scraping
- ▶ **Tools:** Git, GitHub Actions, Linux, Docker, Google Apps Script, Cubic
- ▶ **Typesetting and Visuals:** $\text{\LaTeX}$, TikZ, Manim, Markdown
- ▶ **Web and Databases:** Node.js, Express, Bun, React, Prisma, PostgreSQL, MariaDB
- ▶ **AI Tools:** Claude Code, Codex, OpenCode

LEADERSHIP AND ACTIVITIES

- ▶ Mentored several IUT teams, including IUT_ChronicOverthinkers (later IUT_21stCenturyWitches), best all-girls' team at the ICPC Asia Dhaka Regional 2024 and UIU IUPC 2025
- ▶ UIU CSE Fest 2025: IUPC judge; ICT Olympiad question coordinator under Prof. Nurul Huda 2025
- ▶ Moderator and trainer, UIU Competitive Programming Community 2024 – 2026
- ▶ Programming and decoration teams, IUT Computer Society (IUT 11th National ICT Fest) 2024

LANGUAGES

Bangla (native), English (full professional proficiency)